

Optimizing Bike-Sharing Redistribution in Mexico City: A Machine Learning Approach to Predicting Station Availability in the Ecobici System

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Abstract

This paper presents a data-driven analysis of the Ecobici bike-sharing system in Mexico City, addressing the persistent problem of station depletion through predictive modelling. This research started from a simple everyday frustration: as frequent Ecobici users, the authors often showed up to a station only to find it empty or arrived at a destination and could not return the bike because every dock was taken, so they had to walk at least 3 blocks to find the nearest empty station. Using almost 20,000 station snapshots that were collected over 11 days from 677 stations, we compare linear regression and random forest models for predicting station occupancy. To improve our predictions, we created new variables from the raw data, including cyclic time features, recent occupancy trends, speed changes at each station, historical averages, and grouping the station into geographic zones using K-Means clustering. The Random Forest had better results than Linear Regression, but the difference is not very significant. Based on these results, we propose a two-part redistribution system: one that predicts when stations will run out of bikes before they actually do and another that detects in real time when users open the app and find empty bike docks, triggering an automatic redistribution alert after a predetermined number of Ecobici users send the trigger.

Keywords – bike, sharing, demand prediction, Random Forest, redistribution, urban mobility, unsatisfied demand

I. INTRODUCTION

Bike-sharing systems are a very important part of transportation in big cities worldwide, offering a sustainable and sometimes faster way of travelling, while also helping reduce traffic congestion. Mexico City's EcoBici system is the largest bike-sharing system in Latin America with 9,300 operating bicycles across 698 stations [5]. The system averaged 63,000 trips per day in 2025 [6]. Roughly 67% of users rely on Ecobici to commute to work or school, with an average trip duration of 17 minutes and peak demand on Tuesdays and Thursdays between 7:00-10:00 and 17:00-20:00 hours [6].

Ecobici is also really affordable compared to similar systems in major cities. An annual membership costs 588 Mexican pesos (approximately 28 USD) for unlimited 45-minute rides [7]. In comparison, New York City's Citi Bike charges 239 USD per year for a similar type of service [8], around 8.5 times more expensive. Despite the price difference, Ecobici bikes are used nearly three times more, each bike completes around 9 to 10 trips a day [6], compared to Citi Bike's 3.4 trip per day average [8]. This low cost, high usage has made Ecobici a great alternative to cars and public transit, sometimes even faster than both because of traffic. The system is already expanding for the 2026 FIFA World Cup, with new stations being added near match venues and over 185,000 active memberships as of 2026 [10], making the redistribution challenges addressed in this paper even more urgent.

However, despite its affordability and popularity, the system faces a persistent operational problem: station depletion. This research comes from a real, every day, personal frustration. As an Ecobici user, the author often shows up to a station only to find zero bikes available, especially during rush hours, meaning a walk to the next station, or even having to take the car and taking twice as long. The opposite is just as common a frustrating: arriving at a destination with a bike ready to end the trip only to find the docks full, having to ride around looking for another station. This is not just a minor inconvenience, it makes the whole system less useful and pushes people back towards cars and ride-shares. The fact that this keeps happening after sixteen years suggest that their current redistribution approach is not working, and that is the issue this paper tries to solve.

Previous studies have explored bike-sharing demand prediction using ensemble methods [1], temporal and weather features [2], geographic clustering [3], and fleet optimization models [4].

Right now, Ecobici redistributes bikes using fixed schedules and only reacts after stations are already empty, instead of anticipating the problem. On top of that, there is a huge gap in the data: the system knows how many bikes are at each station, but it has no way of knowing how many people showed up and found no bikes. This paper tackles both problems: first, by building machine learning models that predict when stations will run

out of bike before it actually happens and second, by proposing a new system that captures this missing demand data in real time to trigger automatic redistribution.

III. DATASET AND METHODOLOGY

A. Dataset Description

The dataset used was made up of almost 20,000 station status snapshots collected from the Ecobici GBFS (General Bike share Feed Specification) public API [9] between September 11-21, 2025, covering 677 stations at approximately 10-minute intervals for each data gathering. Each record has a station identifier, geographic coordinates, capacity, available bikes, available docks, operational status and percentage full capacity. A sample of 20,000 records was used for the model training, keeping the proportional distribution across hours and days of the week.

B. Data Preprocessing

Three filtering steps were used to ensure data quality: (1) only operationally active stations were kept, (2) records with missing occupancy values were removed, and (3) records with a discrepancy between station capacity and the sum of available bikes and docks exceeding a tolerance threshold of two units were deleted.

C. Feature Engineering

Six categories of features were developed:

Cyclic temporal encoding. Hours transformed using $\sin(2\pi h/24)$ and $\cos(2\pi h/24)$ to keep the continuity between 23:00 and 00:00.

Log features. Three logs of pct_full values per station provided ~ 10 , 20, and 30 minute occupancy history.

Rate-of-change. Velocity ($\log 1 - \log 2$) and acceleration ($\log 1 - \log 3$) quantified how rapidly stations were getting filled or emptying.

Historical baselines. Mean occupancy per station-hour captured location-specific demand profiles.

Geographic clustering. K-Means ($K=15$, selected with elbow method) grouped stations into logistic zones.

Calendar features. Day of the week and binary weekend indicator.

D. Models

We used two models for comparison. **Linear Regression** was the baseline model, a simple model that tries to find a straight-line relationship between the 14 input features and station occupancy. Since occupancy is percentage, predictions were limited to stay between 0 and 1.

Random Forrest was the main model, made up of 150 decision trees, a maximum depth of 12 levels, at least 5 samples per leaf and square root feature selection at each split. Both models were trained on 80% of the data and tested on the remaining 20%, using R^2 score to measure how well they predicted.

IV. RESULTS

A. Model Performance

Table I compares the results. Random Forest has a significantly better performance than Linear Regression, confirming the non-linear nature of demand patterns.

TABLE I: Model Performance (R^2)

Phase	Lin. Reg.	Rand. Forest
Train	~ 0.76	~ 0.86
Test	~ 0.74	~ 0.76

B. Feature Importance

The most important features were the lag values (pct_full_log1) and historical occupancy, then geographic coordinates and temporal features. This means that the best way to predict a station's occupancy is by looking at what happened there in the last 10–20 minutes and how that specific station usually behaves, instead of just looking at the time of day.

C. When Station Run Out

Station depletion peaks between 14:00-15:00 ($>13\%$ of stations empty), this makes sense because work days frequently end or change to home office around these times. A second peak at 00:00-01:00 shows insufficient overnight redistribution. Lowest depletion happens at 11:00 ($\sim 5\%$). Thursday and Friday have the lowest average availability (~ 6.9 bikes per station), while Sundays show highest (~ 7.7).

Basically, bikes move with the people, in the morning people go to the office and in the afternoon they leave the office. The trucks don't fill the stations fast enough, and at night they stop distributing. This is the whole pattern.

D. Critical Stations

Fifteen stations had more than 26% zero-bike time. Station 552 (Lago Andr maco) has a 100% depletion, this is likely a maintenance issue. Remaining critical stations are concentrated in southern districts like La Del Valle and Narvarte, which is consistent with the models high feature importance with latitude.

V. PROPOSED REDISTRIBUTION SYSTEM

A. Predictive Redistribution

The main idea is simple: instead of waiting for a station to run out of bikes and then reacting, the Random Forrest model predicts which stations will be empty in the next 2 hours, so redistribution trucks can be sent before this even happens.

For this to work, we set two rules based on the data we had. If a station drop below 15% of its capacity, it needs bikes. If it goes above 75%, it has too many. Every hour, we compare the two lists and generate a redistribution plan: move bikes from stations that have too many to stations that are about to run out.

The results from the hourly analysis show clear patterns. On weekdays between 14:00 and 15:00, the number of stations needing bikes peaks, because of office hours. At the same time, stations near office area that are empty in the morning are now full because of commuters. This creates a type of natural flow that the redistribution system can take advantage of. Moving bikes from where there are too many to where they are being taken from.

For example, at 14:00 on a normal weekday, the model shows around 80–100 stations below the 15% limit, while 50–70 others are above 75%. The top critical stations at that hour like, Maria Luz Bringas or Jose Maria Olloque consistently require around 5 to 10 bikes each, while stations in Polanco or Reforma have 10.15 extra bikes that can be redistributed. This kind of hour by hour, station by station plan does not exist in the current system.

There is a similar pattern at night. Between 0:00 and 1:00, over 14% of the stations have zero bikes. Since Ecobici stops lending bikes at 00:30 and reopens at 05:00m any station that empties out in the evening stays empty through the night unless a redistribution truck refills it. This means that early commuters often find stations still empty from the night before.

B. Reactive Layer: Unsatisfied Demand

The predictive model works well for regular patterns, but it cannot anticipate unexpected events like rain, protests, concerts, or holidays. This is where the second layer comes in. The main idea is that the current system has a blind spot: it knows how many bikes are at each station, but it has no way of knowing how many people wanted a bike and could not get one. We propose capturing this missing data through the Ecobici app:

Passive capture: An event is automatically logged with a timestamp, station ID, and location whenever a user opens the app near a station whose bike occupancy falls below 15% or exceeds 75% of its total capacity.

Active capture: An optional button lets users signal they need a bike at their location.

Trigger mechanism: If 10 or more users search for a bike at the same station within 20 minutes, an automated alert is sent to a redistribution truck or dispatch for a truck to go to that station.

VI. CONCLUSION

This paper demonstrated that Random Forest regression with lag features, cyclic encoding, and geographic clustering can effectively predict station occupancy in Mexico City's Ecobici system. We identified clear patterns in when and where stations run out of bikes. The worst hours are between 14:00 and 15:00 on weekdays, when people leave their offices. Thursday and Friday are the hardest days overall. Fifteen stations were empty more than 26% of the time across the entire observation period, and most of them are in the southern part of the city, in areas like la Del Valle and Narvarte.

Based on the patterns we observed, we propose a two part redistribution system. The predictive part uses the trained model to generate hourly redistribution orders, moving bikes from saturated stations to stations that are about to run out. The proposal is to try to not let the stations reach zero, instead get there with the bikes beforehand. The second part is a new concept, that would measure how many users wanted a bike but could not get it, and automatically trigger redistribution when demand spikes unexpectedly. A limitation of this study is the 11-day observation period; future work should have longer time series and include weather data, and event calendars to improve the model.

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